COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 3 — Waterfall Model

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Section 11: Classroom Activity — Choose the Process Model

Comparative analysis of project scenarios to determine the technical and organizational suitability of the sequential Waterfall model.

Scenario 1: College Fee-Reporting Replacement System

- **Suitability Verdict:** Highly Suitable for Waterfall.
- **Engineering Justification:** The operational rules, calculation formulas, user roles, regulatory tax deduction formats, and administrative sign-offs are completely documented, well-understood, and legally fixed. The requirements have minimal volatility, making sequential phase progression (Requirements -> Design -> Implementation -> Verification) efficient, predictable, and low-risk.

Scenario 2: Early-Stage Consumer Social Mobile App

- **Suitability Verdict:** Unsuitable for Waterfall (Requires Agile / Prototyping).
- **Engineering Justification:** The target user behaviors, engagement loops, and market fit are unknown and subject to rapid shifts based on real-world user feedback. In a pure Waterfall model, working software is delivered only at the final phase, creating catastrophic project risk if user assumptions prove false.

Scenario 3: Government Railway Signalling Infrastructure Contract

- **Suitability Verdict:** Highly Suitable for Waterfall (or V-Model variant).
- **Engineering Justification:** Safety-critical public infrastructure demands comprehensive documentation, formal stage-gate audits, strict requirement traceability, and zero mid-cycle scope ambiguity. Strict verification and validation gates at each transition boundary are mandatory for regulatory compliance.

Section 14: Homework — Waterfall Analysis for a Library Management System

Comprehensive architectural evaluation of applying the classical Waterfall model to a centralized Campus Library Management System (LMS).

1. Core System Requirements

- **FR-01 (Book Inventory Search):** The system shall allow users to search titles by ISBN, author name, or subject category.
- **FR-02 (Circulation Issue/Return):** The system shall record book loans against verified student barcode IDs with a fixed 14-day loan period.
- **FR-03 (Automated Fine Computation):** The system shall calculate an overdue penalty of Rs. 5 per calendar day on overdue returns.

- **NFR-01 (Audit Integrity):** All book loan and fine clearance records must be permanently maintained in an append-only audit trail.
- **NFR-02 (Database Concurrency):** The catalog database must prevent simultaneous checkout of the same physical copy by multiple operators.

2. Waterfall Phases and Phase Deliverables

Waterfall Phase	Primary Objective	Mandatory Phase Deliverable
1. Requirements Analysis	Elicit and freeze complete cataloging and circulation rules.	Software Requirements Specification (SRS) signed by Head Librarian.
2. System Design	Define architectural layers, database schema, and ER diagrams.	High-Level & Low-Level Design Document (HLD/LLD).
3. Implementation	Construct database tables, backend business logic, and UI forms.	Tested Source Code & Unit Test Execution Logs.
4. Integration & Testing	Execute comprehensive system, integration, and boundary testing.	Formal Test Summary Report & Defect Matrix.
5. Deployment	Install on campus server environment and migrate legacy physical records.	Production Release Sign-off & Operator Manual.
6. Maintenance	Perform routine database re-indexing, bug patches, and backups.	Maintenance Service Logs & Version Patch Notes.

3. Project Advantages & Limitations

- **Advantage 1: Clear Stage-Gate Governance:** Milestones and delivery schedules are deterministic; administrators know exact documentation and progress status at each stage.
- **Advantage 2: Thorough Engineering Documentation:** Exhaustive design specs ensure that future maintenance personnel can service the system without relying on original developers.
- **Limitation 1: Severe Inflexibility to Change:** If librarians discover that inter-departmental book reservations require distinct workflow rules after coding has commenced, revising the design requires expensive phase rollback.
- **Limitation 2: Delayed Verification:** Working software is invisible to librarians until Phase 4, postponing user experience validation until the end of the project.

4. Change Request Impact Analysis

- **Proposed Change Request (CR-04):** "Introduce automated SMS reminders 2 days prior to book due dates to reduce student delinquency."
- **Impact Assessment:** If submitted during the Implementation phase, this requires revising the SRS (SMS gateway interface, student phone number validation), altering the database schema (notification preference flags, dispatch queue), updating the design document, and modifying test scripts, causing an estimated 10-day schedule slippage.